

# Course Notes

The following notes are intended to remind you what we covered each week. There may be other relevant resources on Blackboard.

## Class 1

This week we covered the following topics:

1. Mathematical Modeling

Instructor note: If you are planning this activity, you need a deck of cards that you can use to divide the class into six groups of four using one characteristic (suit) and then four groups of six using another characteristic (number) so that each group of six has a member from each group of four. You may have more than 24 students, in which case, adapt appropriately!

2. Google NotebookLM

## Mathematical Modeling

We started reading the M3 Mathematical Modeling Materials from the Society for Industrial and Applied Mathematics. You should have a list describing the six stages of making a model.

1. Defining the Problem Statement (Chapter 2)
2. Making Assumptions (Chapter 3)
3. Defining Variables (Chapter 4)
4. Building Solutions (Chapter 5)
5. Analysis and Model Assessment (Chapter 6)
6. Putting it All Together (Chapter 7)

In groups, we read the M3 Challenge Problem and Solution, with each group focusing on a different stage of the modeling process. Then we mixed up into new groups to discuss how those processes fit together into a good overall solution to the challenge problem.

## Google NotebookLM

This is a generative AI tool useful for interacting with documents, especially long ones (we're mathematicians; we read, but we also take productive shortcuts when it helps our learning goals).

We got used to the interface a little bit with the M3 handbook. The link to our notebook is on Blackboard.

**Example 1 Query: What are the six steps involved in mathematical modeling?** *Response:* Mathematical modeling is described in the sources as an iterative process rather than a strictly sequential checklist, meaning that steps are often revisited multiple times as the model is refined. While the guidebook

highlights several key components, the six primary steps involved in the process are as follows:

1. *Defining the Problem Statement.*

Real-world problems are typically broad, complex, and open-ended. The first step involves researching and brainstorming to refine a conceptual idea into a concise problem statement that clearly indicates what the model will predict or measure.

2. *Making Assumptions.*

Because real-world situations often contain incomplete information, assumptions are necessary to simplify the problem and make it tractable. These assumptions help sharpen the focus by reducing the number of factors involved and establishing the specific conditions under which the model is considered valid.

3. *Defining Variables.*

In this stage, the modeler identifies the primary factors influencing the phenomenon and defines them as quantifiable variables with specified units. This includes distinguishing between:

- Independent variables: The model inputs.
- Dependent variables: The model outputs or predictions.
- Model parameters: Quantities that are held constant within the model

4. *Building Solutions (Getting a Solution).*

This step involves using mathematical techniques to generate answers to the defined problem. Depending on the modeler's background and the type of problem, solutions may be reached through hand calculations, algebraic manipulation, computer simulations, or software tools like Excel.

5. *Analysis and Model Assessment.*

After obtaining a solution, the modeler must step back to assess its quality and determine if the results make sense. This involves identifying the model's strengths and weaknesses, as well as performing a sensitivity analysis to see how changes in input values or parameters impact the final output. If the results are unrealistic, the modeler may return to previous steps to adjust assumptions.

6. *Reporting the Results.*

The final step is to explain the model and its implementation clearly so others can use it. This is typically done through a technical report that includes an abstract or executive summary, a clear list of assumptions, defined variables, and an honest assessment of the model's limitations.

Throughout these steps, modelers are encouraged to take careful notes and maintain a bibliography to ensure the process is transparent and repeatable.

*Fact Check:* We decided this was consistent with the guidebook. We can check again by clicking the links to the source material. □

**Example 2 Query: What is the winning paper about?** *Response:* The winning paper, submitted by Team 1356 from Montgomery Blair High School for the 2013 Moody's Mega Math (M3) Challenge, is titled "Analysis of Plastic Waste Production and Recycling Methods". The paper focuses on three main tasks requested by the Environmental Protection Agency (EPA) regarding plastic waste and recycling strategies.

*There is more to see in the Notebook, but that seemed like enough of the response to start with!*

*Fact Check:* We agreed this was a good summary, and we can use the links to the source material to check. □

**Example 3 Query: Why is this a winning paper?** This one is a bit trickier because it is a subjective matter, and we do not have the capacity to ask "How did you come to that conclusion?" and expect an "honest" answer because LLMs are not capable of self-reflection. All that it can do is give a plausible text

response to your question, based on the training data relevant to the question. That is, at best, it's parroting the words of self-reflection!

For comparison, consider a phrase like, "That is a legitimately frustrating experience, and I'm sorry you are feeling bad now." We (people) often use a phrase like this to convey our internal empathy to a listener, who then responds to the perceived internal empathy of the speaker. But I say these words to my toddler all the time, and sometimes I am only *performing* empathy when I don't actually feel it (like at 4 a.m. when he is mad that it's not possible to have a nonconvex triangle in the plane).  $\square$

#### Checkpoint 4 Investigate on your own..

- Navigate to the shared notebook. You can ask questions via the chat!

Professor Gibbons created some Studio Output. What do you think of it?

- Check out the flash cards. Are they useful?
- Are there other outputs you would like the notebook to produce?
- Are there questions you would be unlikely to ask the notebook? Why or why not?

This is just our first foray into using a generative AI tool to enhance our learning. We will develop more skills to interact with NotebookLM throughout the course.

## Class 2

This week we covered the basics of apportionment, plus how to use Google NotebookLM to help dig into primary source material.

### Apportionment

One of the first topics we'll study is **apportionment**.

**Definition 5 Apportionment.** The term **apportionment** refers to the mathematical process of distributing a fixed number of indistinguishable assets among different entities.

Each asset is indistinguishable from any other asset, and no asset can be subdivided.  $\diamond$

**Example 6 Is it apportionment?** For example, suppose Courtney buys 45 boxes of coffee (from the orange and pink doughnut shop down the hill) for the math department to bring to their classes. The department has to come up with a way to decide how many boxes each professor gets, perhaps using the number of students each professor is teaching as the main relevant piece of data to make this decision.

Each professor has to decide how to share the coffee among their classes. This need not be an apportionment problem.

Based on the following table of data, determine how to divvy up the boxes of coffee in different ways. Each box serves 10. Assume every student (and professor) drinks coffee.

**Table 7 Professors and Enrollments, with 45 boxes of coffee.**

| Prof          | Total Students (plus one) |
|---------------|---------------------------|
| Ahmed         | 35                        |
| Bowman        | 60                        |
| Ceniceros     | 71                        |
| Cockburn      | 22                        |
| Dykstra       | 38                        |
| Gibbons       | 55                        |
| Kantrowitz    | 73                        |
| Kuruwita      | 45                        |
| LeMasurier    | 28                        |
| Tripp         | 58                        |
| <i>Totals</i> | 485                       |

Which method or methods seem “fair”? Why? □

One of the most famous apportionment problems is the apportionment of seats in the U.S. House of Representatives among the U.S. states. In fact, the terminology for apportionment problems comes from this application.

**Definition 8 Apportionment Terms.**

|                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Apportionment Problem</b>                                                                                                           | a scenario in which $m$ identical and $n$ indivisible items (the “seats”) must be divided among $n$ entities (the “states”) proportionately (using the “populations”)                                                                                                                                                                               |
| <b>States</b>                                                                                                                          | the entities in the apportionment problem                                                                                                                                                                                                                                                                                                           |
| <b>Seats</b>                                                                                                                           | the items being apportioned among the states                                                                                                                                                                                                                                                                                                        |
| <b>Apportionment Method</b>                                                                                                            | the algorithm determining how to apportion the $m$ seats to the $n$ states, guaranteeing exactly $m$ seats are distributed proportionately to each of the $n$ states based on their populations $p_s$ for $1 \leq s \leq n$ .                                                                                                                       |
| <b>Standard Divisor <math>D</math> and Standard Quota <math>Q_s</math>, Lower Quota <math>L_s</math>, Upper Quota <math>U_s</math></b> | $D = p/m$ where $p = \sum_{s=1}^n p_s$ is the total population; $Q_s = p_s/D$ , representing the “fair” share of seats for state $s$ if the seats could be subdivided; $L_s = \lfloor Q_s \rfloor$ , the largest integer less than or equal to the standard quota; $U_s = L_s + 1$ , the smallest integer strictly greater than the standard quota. |
| <b>Modified Divisor (and friends)</b>                                                                                                  | A value $d$ or $D'$ other than the standard divisor, used to make the distribution of seats produced by rounding (based on a given convention) add up to $m$ . From $d$ each state gets a modified quota $Q'_s$ , lower quota $L'_s$ , and upper quota $U'_s$ .                                                                                     |
| <b>Priority</b>                                                                                                                        | An order by which extra seats are assigned to states. An example is using the fractional part of the quota to determine which states get their upper quota of seats by ranking from largest fractional part to smallest.                                                                                                                            |

◇

**Example 9 Revisiting Coffee.** Taking each professor to be a state and each box of coffee to be a seat, we use the number of students each professor is teaching as the population (plus the professors themselves!). The standard divisor is  $D = \frac{485}{45} = 10 + \frac{35}{45} = 10.\bar{7}$ .

**Table 10 Apportionment Computations**

| Name       | Number of Students (plus one) | Standard Quota | Lower Quota | Upper Quota | Fractional Part |
|------------|-------------------------------|----------------|-------------|-------------|-----------------|
| Ahmed      | 35                            | ?              | ?           | ?           |                 |
| Bowman     | 60                            | ?              | ?           | ?           |                 |
| Ceniceros  | 71                            | ?              | ?           | ?           |                 |
| Cockburn   | 22                            | ?              | ?           | ?           |                 |
| Dykstra    | 38                            | ?              | ?           | ?           |                 |
| Gibbons    | 55                            | ?              | ?           | ?           |                 |
| Kantrowitz | 73                            | ?              | ?           | ?           |                 |
| Kuruwita   | 45                            | ?              | ?           | ?           |                 |
| LeMasurier | 28                            | ?              | ?           | ?           |                 |
| Tripp      | 58                            | ?              | ?           | ?           |                 |

**Solution.**

**Table 11 Standard Quotas**

| Name  | Number of Students (plus one) | Standard Quota | Traditional Round Up or Down | Lower Quota |
|-------|-------------------------------|----------------|------------------------------|-------------|
| A     | 35                            | 3.2474         | 3                            | 3           |
| B     | 60                            | 5.5670         | 6                            | 5           |
| Ce    | 71                            | 6.5876         | 7                            | 6           |
| Co    | 22                            | 2.0412         | 2                            | 2           |
| D     | 38                            | 3.5258         | 4                            | 3           |
| G     | 55                            | 5.1031         | 5                            | 5           |
| Ka    | 73                            | 6.7732         | 7                            | 6           |
| Ku    | 45                            | 4.1753         | 4                            | 4           |
| L     | 28                            | 2.5979         | 3                            | 2           |
| T     | 58                            | 5.3814         | 5                            | 5           |
| Total | 485                           |                | 46                           | 41          |

□

The first apportionment method we'll study is due to none other than Alexander Hamilton!

**Definition 12 Hamilton's Method.** Start by giving each state its lower quota. Then, in order of largest fractional part, distribute the remaining seats. ◇

**Checkpoint 13 The winners are....** Determine which professors get their upper quota of boxes of coffee thanks to Alexander Hamilton.

**Answer.** The professors with fractional rank one through four.

**Table 14 Hamilton Apportionment**

| A | B | Ce | Co | D | G | Ka | Ku | L | T | Total |
|---|---|----|----|---|---|----|----|---|---|-------|
| 3 | 6 | 7  | 2  | 3 | 5 | 7  | 4  | 3 | 5 | 45    |

**Checkpoint 15 A small tweak.** What happens if Courtney only buys 44 boxes of coffee?

Three additional methods involve changing the value of  $D$  to come up with a modified divisor  $d$  where some method (rounding conventionally, using the upper quota, or using the lower quota for each state) adds up to the number of seats. Respectively, these are Webster's method, Adams' Method, and Jefferson's Method.

But what if there were a different way to decide how to round up or down that always added up to the number of seats?

**Definition 16 The Geometric Mean.** The geometric mean of two distinct numbers  $L$  and  $U$  is given by  $c = \sqrt{LU}$ . If  $q > c$ , round up. If  $q < c$ , round down.  $\diamond$

**Definition 17 Huntington–Hill Method (Equal Proportions).** The *Huntington–Hill method* is an apportionment method that assigns seats one at a time using *priority values* based on population and the geometric mean of consecutive seat numbers.

After each state receives one initial seat, additional seats are awarded sequentially to the state with the largest priority value  $\frac{P}{\sqrt{n(n+1)}}$ , where  $P$  is the state’s population and  $n$  is the number of seats the state currently holds.  $\diamond$

**Remark 18** The Huntington–Hill method is sometimes described using different *divisors*, which can be confusing at first. This happens because the method has two equivalent formulations.

In the *priority formulation*, seats are assigned one at a time using the priority value  $\frac{P}{\sqrt{n(n+1)}}$ , where the divisor  $\sqrt{n(n+1)}$  depends on the state’s current number of seats.

In the *rounding formulation*, a single global divisor is used to compute quotas, and each quota is rounded up or down depending on whether it lies above or below the geometric mean  $\sqrt{n(n+1)}$ . Here, the same expression appears as a *rounding threshold* rather than a priority denominator.

Although the divisors look different in these descriptions, both approaches implement the same rule: rounding is determined by the geometric mean, which balances relative over- and under-representation.

**Table 19 Cutoff points for Huntington–Hill rounding of quotas between 1 and 10**

| Quota is between | Cutoff point              |
|------------------|---------------------------|
| 1 and 2          | $\sqrt{2} \approx 1.414$  |
| 2 and 3          | $\sqrt{6} \approx 2.449$  |
| 3 and 4          | $\sqrt{12} \approx 3.464$ |
| 4 and 5          | $\sqrt{20} \approx 4.472$ |
| 5 and 6          | $\sqrt{30} \approx 5.477$ |
| 6 and 7          | $\sqrt{42} \approx 6.481$ |
| 7 and 8          | $\sqrt{56} \approx 7.483$ |
| 8 and 9          | $\sqrt{72} \approx 8.485$ |
| 9 and 10         | $\sqrt{90} \approx 9.487$ |

**Example 20 Huntington–Hill and Parador.**

**Table 21 Huntington–Hill method: Calculations standard divisor  $D = 10,000$**

| State | Population | Standard quota | Cutoff point | Round quota to |
|-------|------------|----------------|--------------|----------------|
| A     | 34,800     | 3.48           | 3.464        | 4              |
| B     | 104,800    | 10.48          | 10.488       | 10             |
| C     | 64,800     | 6.48           | 6.481        | 6              |
| D     | 140,800    | 14.08          | 14.491       | 14             |
| E     | 54,800     | 5.48           | 5.477        | 6              |
| Total | 400,000    | 40             |              | 40             |

**Table 22 Huntington-Hill method: Calculations for with standard divisor  $D = 10,000$  and modified divisor  $d = 10,030$**

| State | Population | Standard quota | Round quota to | Modified quota $d = 10,030$ | Round quota to |
|-------|------------|----------------|----------------|-----------------------------|----------------|
| A     | 34,800     | 3.48           | 4              | 3.47                        | 4              |
| B     | 105,100    | 10.51          | 11             | 10.479                      | 10             |
| C     | 65,100     | 6.51           | 7              | 6.491                       | 7              |
| D     | 140,200    | 14.02          | 14             | 13.978                      | 14             |
| E     | 54,800     | 5.48           | 6              | 5.464                       | 5              |
| Total | 400,000    | 40             | 42             |                             | 40             |

□

## Paradoxes (colloquially)

### Definition 23 Named Paradoxes.

|                           |                                                                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quota paradox</b>      | If there exists an apportionment problem where an apportionment method assigns a state fewer than its lower quota of seats (or more than its upper quota), that method is said to exhibit the quota paradox. |
| <b>Alabama paradox</b>    | This paradox occurs when a state gets apportioned fewer seats simply because the total number of seats being apportioned has increased.                                                                      |
| <b>Population paradox</b> | This paradox occurs when a state $A$ gets apportioned fewer seats than $B$ when the population of $A$ has increased more rapidly than the population of $B$ .                                                |
| <b>New-states paradox</b> | This paradox occurs when a state's apportionment is affected when a new state (together with its fair share of seats) has been added to the computation.                                                     |

◇

**Example 24 Paradox Analysis.** The Intergalactic Congress Apportionments of 2525 and 2535 exhibit one of the paradoxes. Which one?

**Table 25 Intergalactic Congress: Apportionment of 2525**

| Planet   | Population | Standard quota | Lower quota | Surplus | Apportionment |
|----------|------------|----------------|-------------|---------|---------------|
| Alanos   | 150        | 8.3            | 8           | 0       | 8             |
| Betta    | 78         | 4.3            | 4           | 0       | 4             |
| Conii    | 173        | 9.61           | 9           | 1       | 10            |
| Dugos    | 204        | 11.3           | 11          | 0       | 11            |
| Ellisium | 295        | 16.38          | 16          | 1       | 17            |
| Total    | 900        | 50.00          | 48          | 2       | 50            |

**Table 26 Intergalactic Congress: Apportionment of 2535**

| Planet   | Population | Standard quota | Lower quota | Surplus | Apportionment |
|----------|------------|----------------|-------------|---------|---------------|
| Alanos   | 150        | 8.25           | 8           | 0       | 8             |
| Betta    | 78         | 4.29           | 4           | Second  | 5             |
| Conii    | 181        | 9.96           | 9           | First   | 10            |
| Dugos    | 204        | 11.22          | 11          | 0       | 11            |
| Ellisium | 296        | 16.28          | 16          | 0       | 16            |
| Total    | 909        | 50.00          | 48          | 2       | 50            |

□